

THALI

Say what you want. Two robot arms set your table and pour your drink.

Voice in English or Hindi · "stop" works mid-motion · only your voice commands it · runs on the Intel laptop already in the house.

Two SO-101 arms in MuJoCo · Speechmatics realtime · Qwen2-VL on OpenVINO · deterministic safety verifier · SmolVLA / ACT skills

THE PROBLEM

They can still ask. They can no longer reach.

Stroke survivors, people with tremors, older adults living alone: millions who speak perfectly well but cannot lay a table or pour a glass of water without help.

~1.3 billion

people live with significant disability (WHO)

1 in 4

adults will have a stroke; upper-limb weakness is one of the most common lasting effects

Assistive dining today is single-arm, pre-programmed, and expensive

~1.3 billion

people live with significant disability (WHO); 1 in 4 adults will have a stroke, upper-limb weakness among the most common lasting effects

~\$200 + a laptop

two SO-101 arms and the Intel machine already in the house — versus thousands for a single-arm feeding robot

Private by design

planner, verifier and policies run locally on OpenVINO; only speech recognition leaves the device

Same build →

elder care at home · hospital isolation wards · hospitality and kitchen automation; gentle mode when a person is at the table

Next: multi-task SmolVLA fine-tune on the 1050-episode dataset (Kaggle run queued), Core Ultra NPU measurement, real SO-101 hardware.

CLOSED LOOP

Voice → **plan** → **verify** → **act** → **check** → **speak**



Every plan, verdict and skill result is a sha256-chained audit record. Barge-in on partial transcripts pauses the arm within one 20 ms control step.

Measured on Intel Core i7-13650HX CPU + UHD iGPU (no NPU). Same OpenVINO IR runs on a Core Ultra NPU without re-export.

UNDER THE HOOD — that command, from the run's own log

HEARD (operator S1)

"Real estate that we have. Open the top drawer. Pick up the plate with arm A. Place it on the table"

IGNORED (speaker S2, podcast)

"And try things and listen to podcasts and books and...."

PARSED

open the top drawer, put the plate on the table with arm A

PLAN — Qwen2-VL-2B on OpenVINO CPU · source: vlm · verdict: ALLOW

```
[ { "arm": "a", "skill": "open_drawer" },  
  { "arm": "a", "obj": "plate", "skill":  
    "pick_place", "zone": "plate" } ]
```

EXECUTED · oracle ✓ / camera

```
open_drawer(A) ✓  pick_place(A) ✓  →  
success: True
```

AUDIT RECORD sha256

```
63c24a6cce49ee8c36a3cb9f0ecad9f39ddac931...
```

speech-end → plan-ready 16.41 s (VLM on CPU, 4 processes competing) · plan-ready → arms moving 0.078 s · results/demo_seed3.json, results/audit.jsonl

A local VLM proposes. A deterministic verifier decides.

44.6 tok/s

Qwen2-VL-2B INT4 on the Intel CPU · 1130 ms to first token

20/20

unsafe plans blocked · 6/6 sane plans passed

100%

of executed plans verifier-approved (VLM, retry with reasons, then rule fallback)

sha256 chain

tamper-evident audit log — make `verify-log` names the first bad record

WHEN THE PLAN IS UNSAFE, NOTHING MOVES

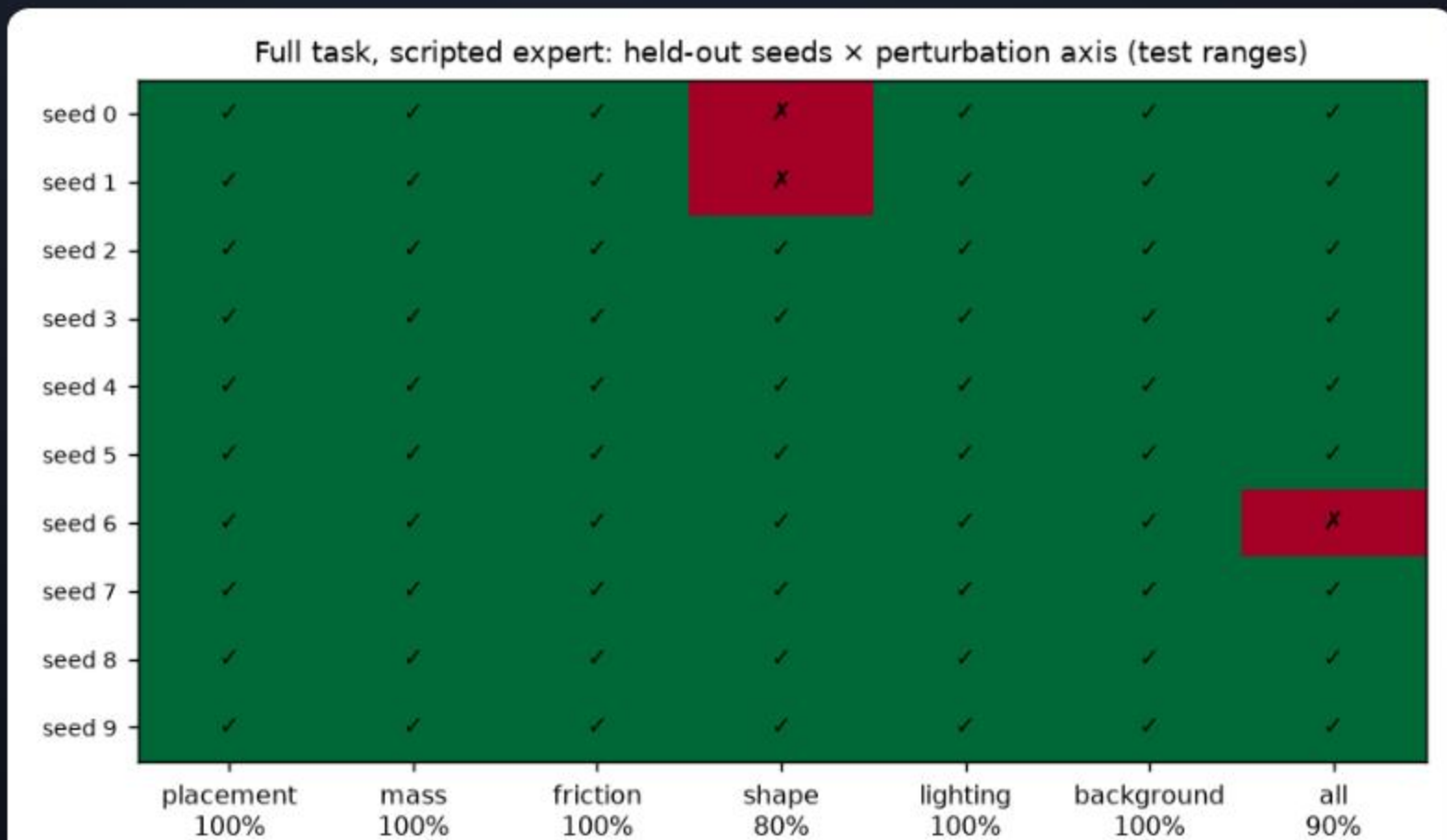
20/20 injected unsafe plans blocked · 6/6 sane plans passed

<code>pour_without_hold</code>	BLOCK	pour by arm a requires the mug held by arm b under the spout
<code>pour_same_arm_holds_mug</code>	BLOCK	mug is out of reach of arm a
<code>pour_after_mug_set_down</code>	BLOCK	pour by arm a requires the mug held by arm b under the spout
<code>handoff_to_self</code>	BLOCK	handoff to the same arm
<code>unknown_object</code>	BLOCK	obj='knife' not in ['plate', 'mug', 'bottle', 'fork_1', 'fork_2', 'spoon_1', 'spoon_2', None]
<code>unknown_zone</code>	BLOCK	zone='sink' not in ['plate', 'fork', 'spoon', 'mug', None]

Fixable ordering mistakes are repaired and returned (REORDER); everything else is refused with the reason spoken back. Every verdict is a sha256-chained audit record — make `verify-log` recomputes the chain.

ROBUSTNESS

Six perturbation axes · held-out test split · 10 seeds each



THE REVERSE TASK

"Clear the table"

9/10

held-out layouts fully cleared: fork back in the drawer, spoon handed B→A and back, drawer shut

2 skills

put_in_drawer · close_drawer — same planner, verifier (close only after every piece is stowed), queues, expert

Cutlery inside the tray is placed on its floor; the piece is carried 6 cm higher over the tray front (its tip would push the drawer shut); the drawer slide has friction like a real one.

"a bit more" · "no, the other side" · "again" · "the other arm"

20/20

scripted corrections resolved to the expected plan against the last executed step

14/16

resolved plans verifier-approved — the rest are refused with a reason (the other arm cannot reach)

A correction never bypasses the verifier: it becomes an ordinary plan first.

BOTH ARMS AT ONCE

Two skills, one physics step barrier

54%

fewer simulation steps: drawer (arm A) and mug (arm B) together — 4/5 sequential → 5/5 concurrent

24%

fewer for plate + mug, same success

Only independent single-arm steps whose objects and targets are ≥ 15 cm apart and outside the shared handoff/pour zone are paired; handoff, hold and pour stay sequential.

LEARNED POLICIES, MEASURED ALONE

Per skill, 20 held-out seeds, policy only

skill (ACT, 50k steps)	success	placement	at 12k steps
open drawer	20/20	—	20/20
hold mug	18/20	—	10/20
plate	15/20	median 1.85 cm	8/20
fork	14/20	median 1.32 cm	6/20
mug	10/20	median 3.92 cm	1/20
handoff · pour	1/20 · 2/20	—	0/20 · 1/20

Training length, not episode count, was the bottleneck (60 → 1050 episodes barely moved it). SmolVLA (multi-task, Kaggle T4 + local): drawer 12/20 at step 9 500, still training.

Static-shape IR → NNCF INT8, behaviour preserved

ACT policy call (batch 1, 3 cameras)	CPU p50	iGPU p50
fp32	49.13 ms	283.24 ms
fp16	49.51 ms	282.82 ms
int8 (NNCF, 300 real frames)	17.06 ms	201.85 ms

Ten test seeds re-run through every precision: identical success — the same skills succeed at fp32, fp16 and int8. Planner: Qwen2-VL-2B INT4 on CPU. SmolVLA vision encoder exported too.

SPEECHMATICS TRACK

Every feature used, every one measured

Realtime + partials	34 partials on an 8 s command; barge-in words matched on partials → arm pauses within one 20 ms step		
End-of-turn	end_of_utterance_silence_trigger = 0.6 s → dispatch on EndOfUtterance; first partial → parsed command 0.45–1.262 s		
Diarization → speaker focus	operator = whoever says "Thali, listen" (or first speaker); noisy.wav's podcast speaker S2 logged and ignored		
Custom dictionary	arm A · arm B · mug · fork · spoon · plate · drawer · Thali		
Hindi	language=hi session → Devanagari → same intents		
TTS loop	"Pouring now. Say stop anytime." · step confirmations · refusals spoken with the reason		
End-to-end latency	speech-end → plan-ready 7.478 s → arms moving 10.447 s (seed 3, noisy sample, VLM on CPU)		
normal.wav (en)	WER 0.00	normalised 0.00	✓ skills recovered
tired.wav (en)	WER 0.44	normalised 0.44	✓ skills recovered
hindi.wav (hi)	WER 1.00	normalised 0.43	✓ skills recovered
noisy.wav (en)	WER 0.31	normalised 0.31	✓ skills recovered

S2 ignored

INTEL BIMANUAL VLA TRACK

What is in the box, and the number behind each row

Task completion + bimanual	7 skills, via-table handoff + hold-while-pour : 9/10 held-out seeds · clear the table 9/10 · both arms at once 54 % fewer sim steps · mid-task recovery 4/4
VLA / multi-modal	1050-episode LeRobot dataset · per-skill ACT at 50k steps: hold 18/20, fork 14/20, plate 15/20 (1–2 cm) · multi-task SmolVLA training · local Qwen2-VL planner 4/8 from the model, 100 % verifier-approved · instruction swap 7/10 · follow-ups 20/20
Robustness	6 axes, held-out split: placement 100 %, mass 100 %, friction 100 %, shape 80 %, lighting 100 %, background 100 %; all six 90 %
OpenVINO / Intel	ACT → IR (static shapes) → NNCF INT8: CPU 49.13 → 17.06 ms ; behaviour identical at every precision; VLM INT4 44.6 tok/s on CPU; Anomalib PatchCore table check AUROC 0.943 on OpenVINO (70.23 ms)
Reproducibility	8 make targets · 80+ tests · CI · Dockerfile · pinned env · dataset on the Hub · README rendered from results/ · live site
Innovation	verifier 20/20 unsafe plans blocked + sha256-chained audit log · speaker-focused barge-in · corrections resolved against the last step

EVIDENCE

What we measured, we show.

Every number in the README is rendered from a JSON file in `results/` by a script in the repo. Held-out seeds. The expert's own ceiling is stated. The learned policies' weak rows are stated. Nothing typed by hand.

9/10

full 7-skill task on held-out seeds (scripted IK expert)

100%

voice samples with the skill sequence recovered (clear / tired / Hindi / noisy)

REPRODUCE · VERIFY · LINKS

Everything is in the repository

github.com/Prashant-thakur77/THALI

code · assets · 75 tests · CI · Dockerfile · make demos / train / eval / bench / demo / verify-log · docs/WRITEUP.md · docs/EVIDENCE.md

huggingface.co/datasets/Prashant-77/thali_all

1050 scripted-expert episodes, 742 837 frames, 3 cameras, LeRobot v3

[video/thali_demo.mp4](#)

4:23 closed-loop demo: live Speechmatics stream, Hindi, full task, pour, blocked plans, barge-in, OpenVINO

[results/*.json](#)

every number on every slide is produced by a script in the repo and rendered from these files

THALI

Two arms. One voice. On the laptop you already own.

github.com/Prashant-thakur77/THALI

huggingface.co/Prashant-77